SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 03 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I Peta Harsha Vardhan 192311039(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

4.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

6. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is –0.6. For an applicant with income = 1.5 and credit score = 2, compute the net input and

apply the step activation function. Determine whether the loan is approved or rejected. (5 Marks)

7. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration. (5 Marks)

8. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical. (5 Marks)

9. A Genetic Algorithm is used to maximize the function $f(x) = 2x + 3$ for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence. (5 Marks)

10. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance. (5 Marks)

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not. (5 Marks)

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration. (5 Marks)

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly. (5 Marks)

14. A Genetic Algorithm is applied to maximize the function $f(x) = 3x + 2$ for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the best candidates. Perform a crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima. (5 Marks)

15. During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Question 1

Question: A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

Concept Involved

Perceptron Model, Net Input (Weighted Sum), Bias and Step Activation Function (Binary Classification).

Definition

A perceptron is the simplest artificial neuron and a linear binary classifier. It computes the weighted sum of its inputs, adds a bias, and passes the result through a step (threshold) activation function that gives an output of 1 or 0.

Net input: $net = (w_1 \times x_1) + (w_2 \times x_2) + b$

Step function: $y = 1$ if $net \geq 0$; $y = 0$ if $net < 0$

Given Data

- Inputs: x_1 = study hours = 2, x_2 = attendance = 1
- Weights: $w_1 = 0.6$, $w_2 = 0.4$
- Bias: $b = –0.5$

Answer (Step-by-Step Solution)

Step 1: Write the formula

$net = (w_1 \times x_1) + (w_2 \times x_2) + b$

Step 2: Substitute the given values

$net = (0.6 \times 2) + (0.4 \times 1) + (–0.5)$

Step 3: Multiply each weight by its input

$0.6 \times 2 = 1.2$

$0.4 \times 1 = 0.4$

$net = 1.2 + 0.4 – 0.5$

Step 4: Add the terms

$1.2 + 0.4 = 1.6$

$1.6 – 0.5 = 1.1$

Net input = 1.1

Feature	Input (x)	Weight (w)	Product (w × x)
Study hours	2	0.6	1.2
Attendance	1	0.4	0.4

Bias	–	–	–0.5
Net input			**1.1**

Step 5: Apply the step activation function

net = 1.1, and $1.1 \geq 0$

Output y = 1

Step 6: Interpret the output (1 = Pass, 0 = Fail)

$y = 1 \rightarrow$ the student passes

Application

The perceptron is applied here as a pass/fail classifier in an educational analytics system. Study hours and attendance are the features, and the weights show that study hours (0.6) influence the result more than attendance (0.4). The same model is used in early-warning systems that identify students at risk so that teachers can give timely support.

Conclusion

The net input is 1.1 and the step function gives an output of 1. Therefore, the student PASSES.

Question 2

Question: In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

Concept Involved

Backpropagation Learning Rule, Delta Rule, Error Calculation, Learning Rate and Weight Update.

Definition

The delta rule (Widrow–Hoff rule) is a gradient-descent learning rule that changes a weight in proportion to the error between the target and actual output, the input value and the learning rate. In backpropagation, this error is propagated backwards through the network so that the weights are updated to reduce the output error.

Error: $e = t - y$

Weight change (delta rule): $\Delta w = \eta \times (t - y) \times x$

New weight: $w(new) = w(old) + \Delta w$

Given Data

- Initial weight: $w(old) = 0.5$
- Learning rate: $\eta = 0.1$

- Input: $x = 2$
- Actual output: $y = 0.6$; target output: $t = 1$

Answer (Step-by-Step Solution)

Step 1: Calculate the error

$e = t - y$

$e = 1 - 0.6$

$e = 0.4$

Step 2: Calculate the weight update using the delta rule

$\Delta w = \eta \times (t - y) \times x$

$\Delta w = 0.1 \times 0.4 \times 2$

$0.1 \times 0.4 = 0.04$

$0.04 \times 2 = 0.08$

$\Delta w = 0.08$

Step 3: Calculate the new weight

$w(new) = w(old) + \Delta w$

$w(new) = 0.5 + 0.08$

$w(new) = 0.58$

Note: the neuron is treated as a linear unit. If a sigmoid unit were used, the derivative $y(1 - y)$ would also be multiplied into the update.

Application

This weight-update procedure is repeated over many iterations and training examples in neural networks used for image recognition, speech recognition and prediction. Each iteration moves the output closer to the target, so the network gradually learns the input–output relationship.

Conclusion

The output error is 0.4, the weight change is +0.08, and the updated weight after one iteration is 0.58. The positive change increases the weight, which moves the next output closer to the target of 1.

Question 3

Question: A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

Concept Involved

Confusion Matrix and Performance Evaluation Metrics (Accuracy, Precision, Sensitivity/Recall, Specificity) of a Deep Learning Model for Disease Prediction.

Definition

A confusion matrix is a table that compares predicted classes with actual classes. It contains True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN), from which the performance metrics are derived.

- **Accuracy** = (TP + TN) / (TP + TN + FP + FN) – overall correctness of the model.
- **Precision** = TP / (TP + FP) – how many predicted positives are truly positive.
- **Sensitivity (Recall)** = TP / (TP + FN) – how many actual patients are detected.
- **Specificity** = TN / (TN + FP) – how many healthy people are correctly identified.

Given Data

	Predicted Positive	**Predicted Negative**	**Row Total**
Actual Positive	TP = 180	FN = 50	230
Actual Negative	FP = 20	TN = 150	170
Column Total	200	200	400

Answer (Step-by-Step Solution)

Step 1: Find the total number of samples

Total = TP + TN + FP + FN = 180 + 150 + 20 + 50 = 400

Step 2: Accuracy

Accuracy = (TP + TN) / Total

Accuracy = (180 + 150) / 400 = 330 / 400

Accuracy = 0.825 = 82.5%

Step 3: Precision

Precision = TP / (TP + FP)

Precision = 180 / (180 + 20) = 180 / 200

Precision = 0.90 = 90%

Step 4: Sensitivity (Recall)

Recall = TP / (TP + FN)

Recall = 180 / (180 + 50) = 180 / 230

Recall = 0.7826 ≈ 78.3%

Step 5: Specificity

Specificity = TN / (TN + FP)

Specificity = 150 / (150 + 20) = 150 / 170

Specificity = 0.8824 ≈ 88.2%

Step 6: Summary of results

Metric	**Formula**	**Value**
Accuracy	(TP + TN) / Total	82.5%
Precision	TP / (TP + FP)	90.0%

Sensitivity (Recall)	TP / (TP + FN)	78.3%
Specificity	TN / (TN + FP)	88.2%

Discussion – Reliability for Medical Diagnosis

The model has good precision (90%) and specificity (88.2%), so it gives few false alarms. However, its recall is only 78.3%. The proportion of patients that the model misses is:

$$FN / (TP + FN) = 50 / 230 = 0.217 = 21.7\%$$

So about 22 out of every 100 patients are wrongly classified as healthy. In medical diagnosis a false negative is more dangerous than a false positive, because a missed disease delays treatment and can be life-threatening, whereas a false positive only leads to further tests.

Application

These metrics are used to evaluate deep-learning systems for cancer, diabetes and COVID-19 screening and for medical image analysis. They help doctors decide whether a model can be deployed and whether the decision threshold should be tuned to increase recall.

Conclusion

Accuracy = 82.5%, Precision = 90%, Recall = 78.3% and Specificity = 88.2%. Because 21.7% of actual patients are missed, the model is NOT sufficiently reliable to be used alone for medical diagnosis. It can serve as a screening or decision-support tool, but it should be improved (more data, class balancing, threshold tuning) to raise recall and be confirmed by clinical tests before use.

Question 4

Question: A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Concept Involved

Genetic Algorithm, Population, Fitness Function, Selection, Crossover and Mutation (Hypothesis Space Search).

Definition

A Genetic Algorithm (GA) is a search and optimisation technique inspired by natural evolution. A population of candidate solutions (chromosomes) is evaluated by a fitness function; the fittest are selected, combined by crossover to form offspring, and randomly altered by mutation to produce better solutions over generations.

Given Data

- Objective: maximise $f(x) = x^2$
- Initial population: 2, 4, 6, 8, represented as 4-bit binary strings (bit values 8, 4, 2, 1)

Answer (Step-by-Step Solution)

Step 1: Encode each individual in binary

$2 = (0\times8) + (0\times4) + (1\times2) + (0\times1) = 0010$

$4 = (0\times8) + (1\times4) + (0\times2) + (0\times1) = 0100$

$6 = (0\times8) + (1\times4) + (1\times2) + (0\times1) = 0110$

$8 = (1\times8) + (0\times4) + (0\times2) + (0\times1) = 1000$

Step 2: Calculate the fitness using $f(x) = x^2$

$f(2) = 2^2 = 4$

$f(4) = 4^2 = 16$

$f(6) = 6^2 = 36$

$f(8) = 8^2 = 64$

Sum of fitness = 4 + 16 + 36 + 64 = 120

Individual (x)	Binary	Fitness x^2	Share of total (fitness / 120)	Rank
2	0010	4	4/120 = 3.3%	4
4	0100	16	16/120 = 13.3%	3
6	0110	36	36/120 = 30.0%	2
8	1000	64	64/120 = 53.3%	1

Step 3: Selection

The two highest fitness values are 64 and 36.

Selected parents: x = 8 (1000) and x = 6 (0110)

Step 4: Crossover by exchanging the least significant bits (LSB)

Parent 1 = 100[0] LSB = 0

Parent 2 = 011[0] LSB = 0

Swapping the LSBs ($0 \leftrightarrow 0$) gives: Offspring 1 = 1000 = 8, Offspring 2 = 0110 = 6

Since both parents have LSB = 0 (all values in the population are even), exchanging only the LSB gives offspring identical to the parents. To obtain new offspring, the crossover is extended to the last two bits (crossover point after the 2nd bit):

Parent 1 = 10|00 Parent 2 = 01|10

Offspring 1 = 10|10 = $(1\times8) + (0\times4) + (1\times2) + (0\times1) = 10$

Offspring 2 = 01|00 = $(0\times8) + (1\times4) + (0\times2) + (0\times1) = 4$

Step 5: Fitness of the offspring

$f(10) = 10^2 = 100$ (greater than 64 and 36, so it is better than both parents)

$f(4) = 4^2 = 16$

Role of Mutation

Mutation randomly flips one or more bits of a chromosome with a small probability. Its roles are:

- **Maintains diversity:** it introduces new genetic material that crossover cannot create. For example, flipping the LSB of 1000 gives 1001 = 9, with fitness $9^2 = 81$.
- **Prevents premature convergence:** it stops the population from becoming identical and getting trapped in a local optimum.
- **Improves effectiveness:** it lets the search explore new regions of the hypothesis space, so the global maximum can be reached.

Application

Genetic algorithms are applied in function optimisation, neural-network weight and architecture tuning, feature selection, scheduling, route planning and engineering design problems where gradient-based methods are difficult to use.

Conclusion

The fitness values are 4, 16, 36 and 64, so the best candidates are 8 and 6. An LSB-only crossover leaves them unchanged because both end in 0, while a two-bit crossover produces x = 10 with a higher fitness of 100. Mutation is essential to preserve diversity and help the GA reach the optimum.

Question 5

Question: During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model.

Concept Involved

Overfitting in Deep Neural Networks, Generalisation, Training vs Validation Accuracy, Regularisation and Early Stopping.

Definition

Overfitting occurs when a model learns the training data too closely, including its noise and random details, so it performs very well on training data but poorly on unseen data. Validation accuracy is the accuracy measured on a separate dataset that is not used for weight updates; it estimates how well the model generalises.

Given Data

- Training accuracy: 60% (start) → 95% (end), increasing steadily.
- Validation accuracy: 58% (start) → 72% (peak) → 68% (later epochs).

Analysis of the Trend

Step 1: Calculate the gap between training and validation accuracy

Gap at start = 60 – 58 = 2 percentage points

Gap at end = 95 – 68 = 27 percentage points

Increase in gap = 27 – 2 = 25 percentage points

Step 2: Calculate the fall in validation accuracy after the peak

72 – 68 = 4 percentage points

Stage	Training accuracy	Validation accuracy	Gap
Start of training	60%	58%	2
End of training	95%	68%	27

Up to the point where validation accuracy peaks (72%), both curves improve, so the model is learning useful general patterns. After this point, training accuracy keeps rising while validation accuracy drops by 4 points. This diverging trend and the widening gap are the typical signs of OVERFITTING. The best model is the one at the peak of validation accuracy.

Why Overfitting Occurs

- The network has high capacity (many layers and parameters) compared with the amount of training data.
- Training continues for too many epochs, so the model starts memorising noise and specific training examples.
- The training data is small, noisy or not representative, and no regularisation is applied.

Techniques to Reduce Overfitting

- **1. Early stopping:** monitor the validation accuracy or loss and stop training (restoring the best weights) when it no longer improves, here near the 72% peak.
- **2. Regularisation / Dropout:** L2 weight decay penalises large weights, and dropout randomly switches off neurons during training so that the network cannot depend on specific neurons. This forces it to learn robust features.

Other helpful methods are data augmentation, collecting more data and reducing the model size.

Significance of Validation Accuracy

- It measures performance on unseen data, giving a realistic estimate of generalisation rather than memorisation.
- It helps to detect overfitting or underfitting during training.
- It is used for hyperparameter tuning, model selection and deciding when to stop training.
- The final, untouched test set is used only once for the final evaluation.

Application

Monitoring training and validation curves is standard practice in deep-learning applications such as image classification, speech recognition, fraud detection and medical diagnosis, to ensure that the deployed model works reliably on new real-world data.

Conclusion

The model suffers from overfitting after the validation accuracy peaks at about 72%. It memorises the training data (95% accuracy) while its performance on unseen data falls to 68%, a gap of 27 points. Applying early stopping and regularisation (dropout/L2) improves generalisation, and validation accuracy is the key metric for judging the true performance of the model.